

SDS 192 Midterm Exam

Professor Lindsay Poirier

Name:

Instructions:

- This midterm lasts 75 minutes.
- This exam is closed-book, to be individually completed and without the aid of the internet or mobile phones. You may reference the ‘R’ cheatsheets, along with any supplementary material I have provided.
- If there are any areas of the exam that are ambiguous to you, please note that on the exam, state your assumptions, and use your best judgment.

Several questions in this exam will reference an example dataset. That dataset is the 2018 Central Park Squirrel Census. I have provided you with the documentation for that dataset. Please note that you will be working specifically with the **Hectare Data**.

1. What is the unit of observation in the example dataset?
2. Identify one nominal, one ordinal, one discrete numeric variable in the example dataset.

3. What are the characteristics of a data frame in R? What are the characteristics of a vector? How are they related?

4. What is the chief difference between a barplot and a histogram?

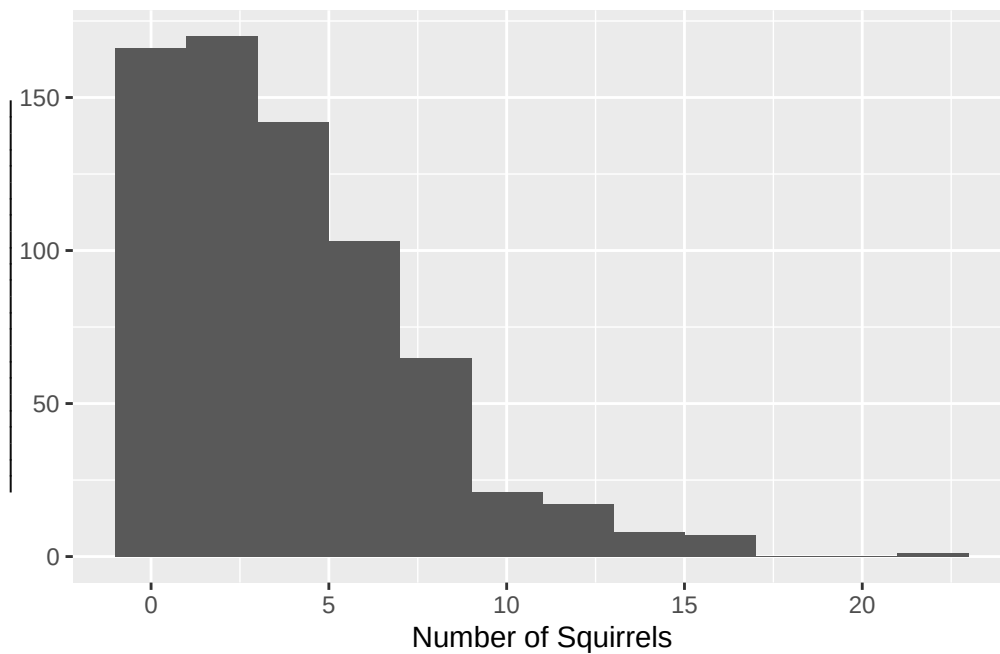
5. We saw in class that there are two ways to create a barplot using `ggplot`: either using `geom_bar()` or using `geom_col()`. Illustrate the need for these two different ways using simple data frame examples.

6. What is overplotting? What are two ways that we can deal with overplotting?

7. When would we use each of the following types of color palettes on a plot: a qualitative palette, a sequential palette, and a divergent palette?

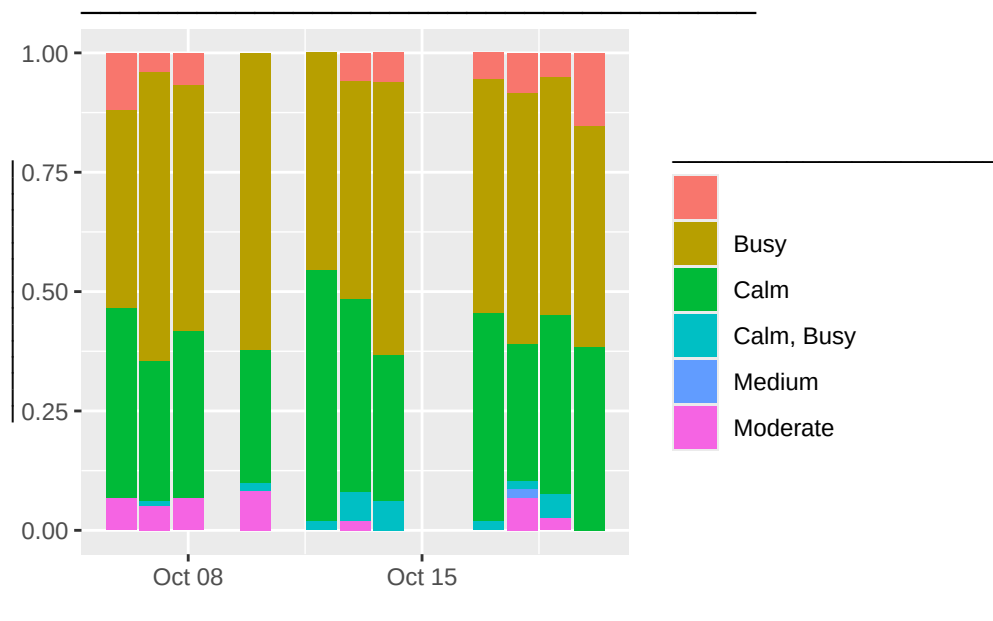
8. Given the example dataset, what would be an appropriate y-axis label for the following plot?

```
nyc_squirrel_census |>
  ggplot(aes(x = number_of_squirrels)) +
  geom_histogram(binwidth = 2) +
  labs(x = "Number of Squirrels",
       y = "_____")
```



9. Develop a descriptive title and labels for the following plot.

```
library(lubridate)
nyc_squirrel_census |>
  mutate(date = mdy(date)) |>
  ggplot(aes(x = date, fill = hectare_conditions)) +
  geom_bar(position = "fill") +
  labs(title = "_____ ",
       x = "_____ ",
       y = "_____ ",
       fill = "_____ ")
```

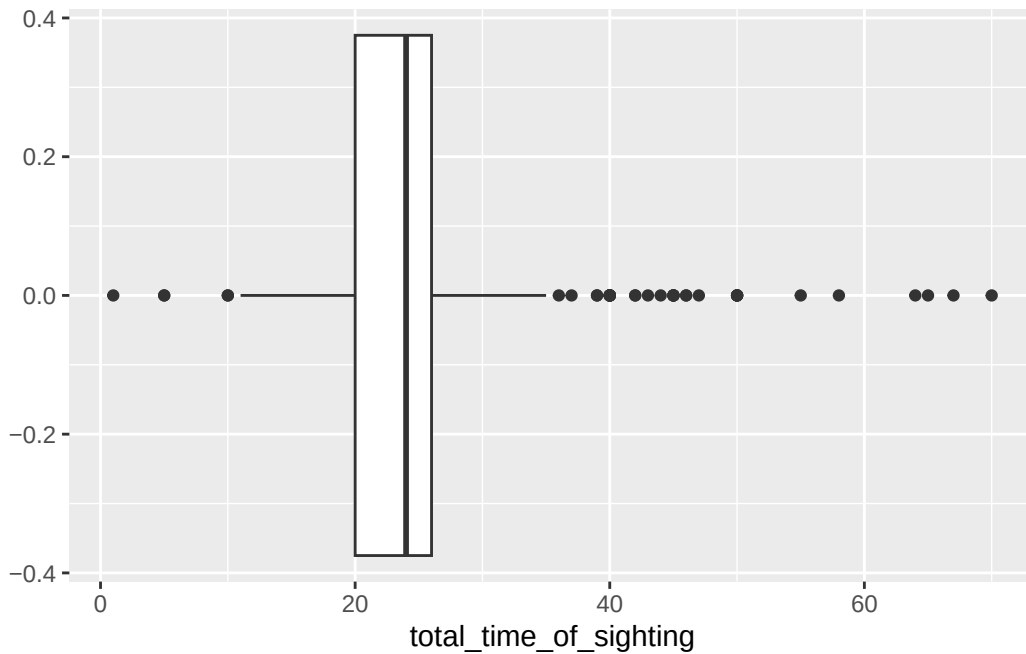


10. What is the scale of the x-axis on the previous plot? What is the scale of the y-axis?

11. According to Tufte, what is a plot's lie factor?

12. Approximately what is the interquartile range represented on the following plot?

```
nyc_squirrel_census |>
  ggplot(aes(x = total_time_of_sighting)) +
  geom_boxplot()
```



13. Two students were working on the same lab assignment and tried to edit the same line in the file. The merge conflict below appeared. What should Student 2 do to ensure that their version of the code gets pushed to the remote repository?

```
<<<<<<< HEAD:lab1.Rmd
#Student 1's code
ggplot(lobby_chicago, aes(x = ACTION)) + geom_bar()
-----
#Student 2's code
```

```
ggplot(lobby_chicago, aes(x = PURPOSE)) + geom_bar()
>>>>>> hash
```

14. Fill in the following statement with the correct `git` vocabulary.

`git` _____ downloads a remote repository to your local computer. `git` _____ fetches changes to code from a remote repository to your local machine. _____ involves selecting files we wish to commit changes on. `git` _____ saves a snapshot of the current version of code to the git version control. `git` _____ uploads changes to a remote repository.

15. What is branching? Outline a collaborative GitHub workflow that engages branching to incorporate multiple changes into a final document.

16. In what hectares and on what dates and shifts did anonymous sighter 37 count? Write pseudocode to produce the following data frame.

	hectare	shift	date	anonymized_sighter
1	11G	AM	10122018	37
2	11H	AM	10102018	37
3	14G	AM	10122018	37
4	16G	AM	10102018	37
5	18I	AM	10172018	37
6	21H	AM	10172018	37
7	21I	AM	10182018	37
8	22H	AM	10182018	37

9	38I	AM 10192018	37
10	39I	AM 10192018	37

17. What would be the difference in outputs between the following two code chunks?

```
mean(nyc_squirrel_census$total_time_of_sighting, na.rm = TRUE)

nyc_squirrel_census |>
  summarize(mean_time_of_sighting = mean(total_time_of_sighting, na.rm = TRUE))
```

18. In which 10 counting sessions where the most squirrels sighted per minute? Return for me their hectares, shifts, dates, and sightings per minute. Write pseudo-code to produce the following data frame.

	hectare	shift	date	sightings_per_minute
1	35F	AM	10072018	1.0000000

2	04C	PM 10102018	0.8500000
3	07F	AM 10082018	0.7727273
4	01B	AM 10122018	0.7391304
5	03B	PM 10072018	0.7333333
6	12F	PM 10072018	0.7142857
7	20F	PM 10072018	0.7000000
8	32E	AM 10072018	0.7000000
9	05C	PM 10102018	0.6500000
10	33E	AM 10142018	0.6500000

19. In which 5 hectares were the most total squirrels sighted? How much time was spent sighting in each of these hectares? Write pseudo-code to produce the following data frame.

```
# A tibble: 5 x 3
  hectare total_squirrels total_time_hectare
  <chr>          <int>          <int>
1 14D             32             90
2 32E             30             45
3 14E             28             65
4 01B             27             58
5 07H             26             55
```

20. What was the total, mean, median time spent sighting squirrels in AM vs. PM counting sessions, and how many times was the total time spending counting recorded as NA? Write pseduo-code to produce the following data frame.

```
# A tibble: 2 x 5
  shift total_time mean_time median_time na_time
  <chr>      <int>      <dbl>      <int>    <int>
1 AM         8457        24.9         24        11
2 PM         8480        24.9         24         9
```